

total binding energy for X and Y is larger.

30. **Ans: D**

Initial **actual** count rate $C_0 = 39.0 - 5.0 = 34.0 \text{ s}^{-1}$

After 20 mins = 2 half-lives, the count rate becomes $34.0 \times (1/2)^2 = 8.5 \text{ s}^{-1}$

Since radioactive decay is random and the sample is a point source, then the count rate will be inversely proportional to the square of the distance from source to detector, ie.

$$C \propto \frac{1}{r^2}$$
$$\frac{C'}{C} = \frac{\frac{1}{20^2}}{\frac{1}{40^2}} = 4$$

That means the count rate at 20 cm from the source would measure 4 times the count rate at 40 cm from source. Thus the count rate at the 20 cm position would read $8.5 \times 4 = 34.0 \text{ s}^{-1}$

The **measured** count rate at the 20 cm position would be $34.0 + 5.0 = 39.0 \text{ s}^{-1}$